

Data Preparation and Visualization Report

Rental Affordability across Portugal's NUTS III Regions in 2024

Which Portuguese regions are least affordable to rent in on a local salary, and what do they have in common?

Team T06

Kevin Gomez	202602766
Lara Ribeiro	202306672
Maria Benedita Marques	202306688

Course lecturer

Professor José Luís Moura Borges

30/10/2026

Contents

1 Introduction and research question **1**

2 Data and preprocessing **1**

2.1 Indicators 1

3 Visualization design **2**

3.1 Composition and visual hierarchy 3

3.2 Geographical representation and house indicators 3

3.3 Classification and consistency across views 3

3.4 Colour and typography 4

3.5 Contextual dot plots 4

3.6 Labels, annotations, and readability 5

3.7 Design trade-offs 5

4 Design evolution and validation **5**

4.1 Evolution and validation of the geographical visualization 6

4.2 Development of the complementary visualizations 7

5 Final visualization and findings **8**

5.1 Regional differences in rental affordability 8

5.2 Shared characteristics and contrasting profiles 8

6 Limitations and conclusion **9**

6.1 Limitations 9

6.2 Conclusion 9

7 Team contributions and use of generative AI **10**

7.1 Team contributions 10

7.2 Use of generative AI 10

8 References **11**

1 Introduction and research question

Housing affordability has become a prominent topic of public debate in Portugal. Rising housing prices and rents have drawn attention to the difficulties of accessing suitable accommodation and to differences in affordability across the country. The social relevance of this issue motivated our choice of topic and our interest in examining how rental costs compare with local earnings.

Rent levels alone, however, do not provide a complete measure of affordability. A region with higher rents may also have higher earnings, while relatively low rents can still represent a substantial financial commitment where earnings are lower. Assessing rental costs in relation to earnings therefore provides a more informative basis for comparing regions.

This project examines rental affordability across Portugal's 26 NUTS III regions in 2024. It addresses the following research question:

> *Which Portuguese regions are least affordable to rent in on a local salary, and what do they have in common?*

To support a consistent comparison, we consider a reference dwelling of 80 m^2. Its estimated monthly rent is calculated using the regional median rent per square metre of new rental contracts and expressed as a percentage of regional mean gross monthly earnings. The fixed dwelling size enables comparisons under a common assumption: the resulting indicator does not represent the actual housing expenditure of individual households.

The final visualization explores four contextual characteristics: resident population, the share of foreign residents, tourist nights per resident, and newly completed dwellings. These variables contextualize regions with higher rent-to-earnings ratios through demographic concentration, international population presence, tourism intensity, and housing construction.

The project communicates these comparisons through a coordinated visualization combining a regional map, proportional house indicators, and contextual dot plots. Its purpose is to make regional differences accessible to non-specialist readers and identify associations that help contextualize rental affordability, without interpreting those associations as evidence of causation.

2 Data and preprocessing

We used data from Statistics Portugal (INE) and the Agency for Integration, Migration and Asylum (AIMA) [1, 5]. All regional comparisons use the 26 NUTS III regions of Portugal under the NUTS 2024 classification, including the Azores and Madeira. The map uses official boundaries from Eurostat/GISCO [2].

The INE files were exported as CSV files with introductory text, multi-level headers, and decimal commas, so we kept only the rows with a region code, converted the values to numbers and reshaped each table to one row per region and year. Monthly tourist nights were summed by year. We joined all tables by region code and year, and the figure uses only 2024. Before drawing, the notebooks checks that all the columns exist, that there is one row per region, and that population, earnings and rent have valid values. We also compared some values (Porto Metropolitan Area, Greater Lisbon and the Algarve) with the INE files.

2.1 Indicators

Table 1 summarizes how each indicator in the figure was calculated and where they come from. All indicators refer to 2024.

Indicator	Calculation	Source and year
Rent-to-earnings ratio (%)	Median rent per m^2 × 80 ÷ mean gross monthly earnings × 100	INE, 2024
Resident population (million)	Resident population ÷ 1,000,000	INE, 2024
Foreign residents (% of population)	Foreign residents ÷ resident population × 100	AIMA, 2024; INE (population)
Tourist nights per resident	Tourist nights in the 12 months ÷ resident population	INE, 2024
New homes completed	Homes completed in new buildings for families (number)	INE, 2024

Table 1. Definitions and sources of the indicators, 2024.

The rent-to-earnings ratio is the main indicator:

$$R_i = \frac{80\, r_i}{e_i} \times 100, \tag{1}$$

where r_i is the median monthly rent per square metre of new rental contracts in region i, and e_i is regional mean gross monthly earnings.

For example, in the Porto Metropolitan Area, the median rent is €9.14 per m^2, so the estimated monthly rent of the 80m^2 is €731, wich represents 45.1% of mean gross monthly earnings (€1,620.10). The four regions where this ratio exceeds 50% are highlighted in the figure.

The other four indicators describe what these regions have in common. Foreign residents and tourist nights are divided by the population, so regions of different sizes can be compared. Population and new homes completed are shown as totals.

3 Visualization design

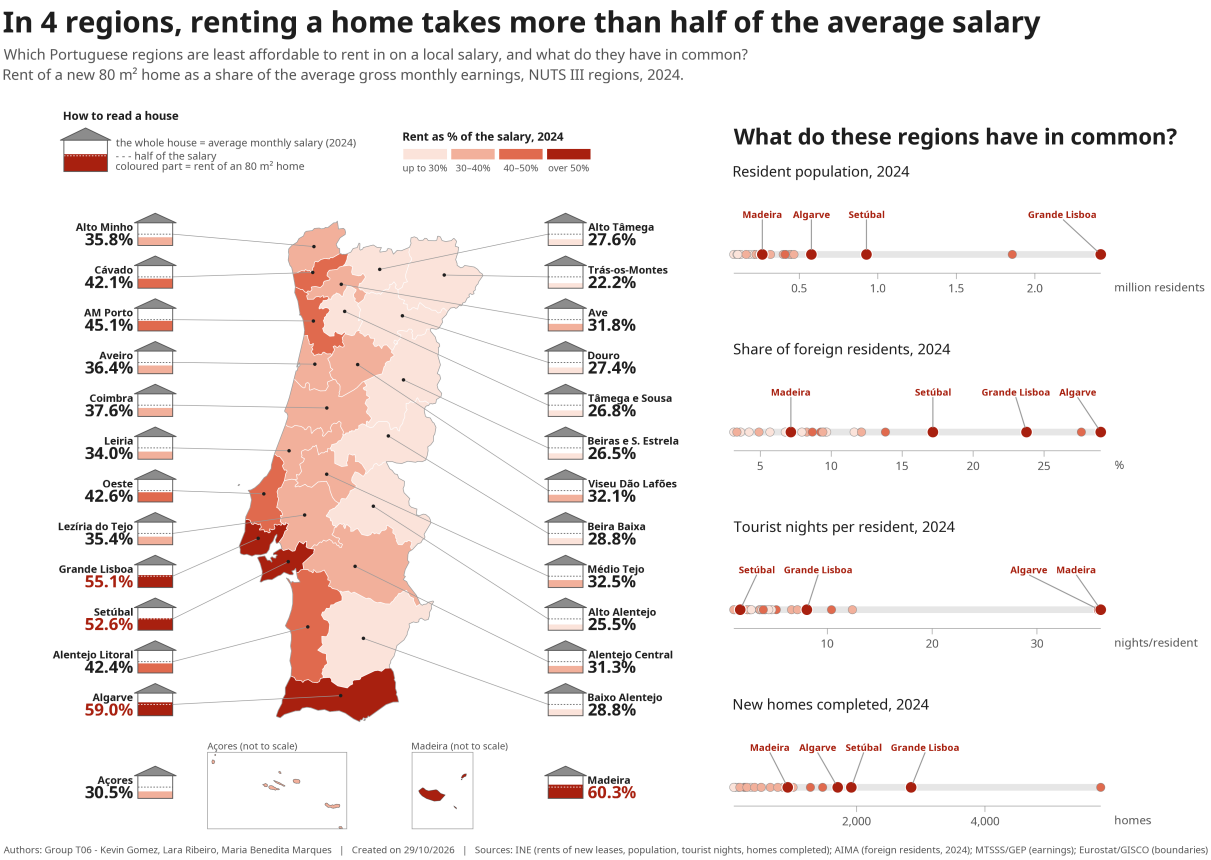


Figure 1. Rental affordability and regional context across Portugal's NUTS III regions, 2024.

The visualization was designed to help non-specialist readers understand how rental affordability varies across Portugal and explore the characteristics of regions where estimated rent represents a larger share of earnings. To support these connected tasks, it combines a geographical map, proportional house indicators, and four contextual dot plots. The map establishes where the regions are located, the houses communicate their rent-to-earnings ratios, and the dot plots place the highlighted regions within broader regional distributions. Together, these components support geographical orientation, numerical comparison, and contextual exploration.

3.1 Composition and visual hierarchy

The composition places the map and house indicators on the left and the contextual panels on the right. Greater space is allocated to the geographical component because it contains the regional boundaries, connecting lines, and 26 labelled house symbols. The right-hand panels extend this overview by showing how the regions compare across demographic, tourism, and construction measures.

This arrangement allows readers to move between complementary views without consulting separate figures. The same affordability colours are repeated throughout the composition, while direct labels identify the highlighted regions. Readers can therefore recognize corresponding groups across panels without having to learn a different encoding for each view.

The main title introduces the finding that four regions exceed a rent-to-earnings ratio of 50%, while the subtitle specifies the year, geographical coverage, and reference dwelling. Above the map, an example house and a colour legend explain the encodings before readers examine individual regions. The heading above the contextual panels introduces the question guiding the comparison of the highlighted cases.

3.2 Geographical representation and house indicators

We selected a choropleth map to provide geographical context and reveal broad territorial differences in rental affordability. Portugal's recognizable outline also acts as a visual entry point, encouraging readers to locate familiar areas and explore their values. However, because larger regions occupy more space regardless of their affordability ratio, the map is complemented by equally sized house indicators and numerical labels.

Regional names and houses are arranged in two columns outside the mainland map, leaving smaller geographical areas unobstructed. Points within the regions and connecting lines establish the correspondence between each territory and its indicator. This uses the Gestalt principle of connectedness: visibly linked elements are perceived as belonging together. The approximately north-to-south arrangement further supports geographical orientation. Açores and Madeira are included in separate insets, explicitly labelled as not being to scale.

Each house has an identically sized rectangular body representing 100% of regional mean gross monthly earnings. Within that body, the filled height represents the estimated monthly rent of the reference $80\,\mathrm{m}^2$ dwelling as a percentage of those earnings. Consequently, differences in the filled proportions reflect differences in the indicator rather than changes in the dimensions of the symbols.

The roof completes the housing metaphor, making the subject recognizable and helping attract attention, while the quantitative scale remains confined to the rectangular body. A dashed horizontal line marks 50% of earnings, providing a shared reference for assessing whether estimated rent exceeds half of the regional mean. The exact percentage beside each house supports precise reading, with values above 50% additionally emphasized in dark red.

This combination balances engagement and precision. Although the houses make the subject visually accessible, their different vertical positions make small differences less straightforward to compare than they would be in an aligned bar chart. The numerical labels allow readers to check values directly.

3.3 Classification and consistency across views

The four affordability classes are defined as up to 30%, above 30% up to 40%, above 40% up to 50%, and above 50%. These round boundaries provide a simple structure for comparison, with the final

threshold corresponding directly to half of mean gross monthly earnings. They serve as presentation choices rather than official classifications of housing risk.

Using discrete classes makes broad differences easier to identify, but also means that distinct values within a class receive the same colour. For this reason, colour is complemented by proportional house fills and exact percentages. These redundant encodings support different levels of reading, from an initial overview to closer numerical comparison.

The same classification is repeated in the map, houses, and contextual dots. Through the Gestalt principle of similarity, this repetition visually connects regions belonging to the same affordability class. Roofs, boundaries, and connecting lines use neutral greys so that supporting elements remain less prominent than the information being compared.

3.4 Colour and typography

Since the ratio goes from low to high, we used a sequential palette with four classes, from light to dark red. We made it by hand with one hue, and each class is clearly darker than the one before (`#FBE3DA`, `#F2B09B`, `#E06A4E`, and `#A8200F`). The limits are 30%, 40%, and 50%, so the last class matches the 50%.Only the regions above 50% use the darkest red, so it works as the accent.

We checked the whole figure in the HCL Wizard colour-vision emulator [4]. In grayscale and in three types of colour blindness, the classes keep their order and the regions above 50% stay the darkest. A thin grey outline keeps the lightest class visible. .

All text in the visualization uses Noto Sans. The Nightingale guide (France, 2021) asks for a sans-serif font with a large x-height and tabular numbers [3]. We compared four fonts and measured the letter x and the digits. We kept Noto Sans: its x-height is large (only Verdana's is bigger), its numbers are tabular, and the guide recommends it for projects with several languages. Size and weight give the hierarchy, and no text is smaller than 9 pt.

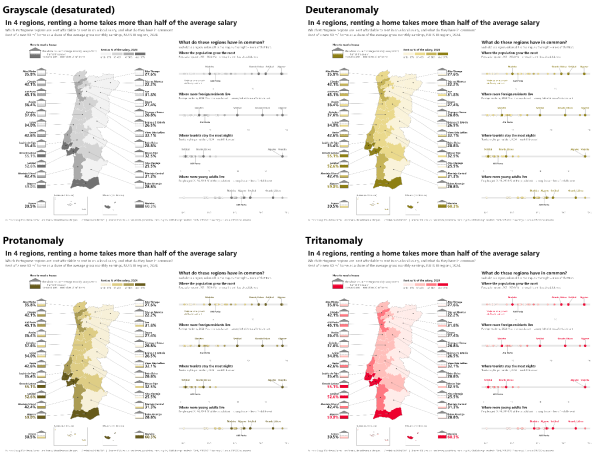


Figure 2. Colour-vision simulations: grayscale and three colour-vision deficiencies.

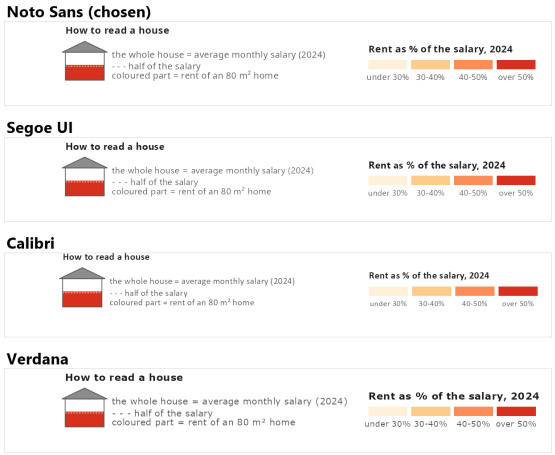


Figure 3. Examples from the typeface comparison.

3.5 Contextual dot plots

The four contextual panels show resident population, the foreign-resident share, tourist nights per resident, and newly completed dwellings in 2024. We selected horizontal dot plots to display the regional distributions compactly while retaining individual observations. Within each panel, horizontal position represents the contextual value, taking advantage of the accuracy with which readers can compare positions along a common scale.

Because the characteristics have different meanings and magnitudes, each panel has its own scale and units. Population is presented in millions of residents, foreign residents as a percentage of the population, tourism as nights per resident, and construction as the number of newly completed dwellings. Numerical labels and tick marks support value estimation, with units placed alongside the numbers.

Accordingly, horizontal positions should be compared within each panel rather than directly across panels.

Point colours connect these distributions to the affordability classes shown in the map. Grande Lisboa, Península de Setúbal (labelled Setúbal in the figure), Algarve, and Madeira—the four regions above 50%—are also directly labelled and represented by larger points. This additional size distinction highlights membership of the selected group; it does not proportionally encode population or any other numerical measure. Connecting lines preserve the association between labels and points when labels are displaced to avoid overlap.

The remaining regions provide context for assessing whether highlighted cases lie near the centre or extremes of each distribution, although overlapping points can obscure individual observations. The panels deliberately omit statements about association strength, allowing readers to examine the patterns before consulting the report's interpretation.

The measures nevertheless require different readings. Population and construction are absolute counts, whereas the foreign-resident share and tourist nights per resident account for population size. The construction panel therefore shows where more dwellings were completed, rather than whether construction was sufficient relative to regional population or housing demand.

3.6 Labels, annotations, and readability

Text formatting distinguishes the overall message, panel titles, regional values, and supporting information. Exact affordability percentages are emphasized beside the houses, while the individual dot-plot titles use smaller, regular-weight black text to remain readable without competing with the main headings. Axis numbers, units, and explanatory labels use subdued grey text.

Direct labels reduce the need to repeatedly consult the legend or trace colours between views. Within the map component, regional names sit beside their corresponding houses; within the contextual panels, labels identify the four highlighted regions. Explanatory annotations are concentrated in the example house, colour legend, and island inset labels, where they clarify how to interpret the figure.

White space separates the main components and contextual panels, while the proximity of titles, labels, and axes helps readers perceive each panel as a coherent group. Restrained connecting lines maintain the necessary links without dominating the composition.

3.7 Design trade-offs

The combined representation reflects a trade-off between different analytical tasks. An ordered bar chart would facilitate ranking and precise comparison of affordability values, but would provide less immediate geographical context. Conversely, a map alone would offer limited support for reading exact values or examining contextual characteristics. Bringing the map and houses together addresses these complementary needs.

Similarly, scatter plots would show relationships between affordability and each characteristic more directly through two positional axes. The selected dot plots instead prioritize regional distributions and the identification of highlighted cases, with affordability represented through colour classes. This provides a compact contextual comparison, while concealing differences within each affordability class.

The design's creative contribution lies in coordinating a recognizable housing metaphor with geographical and statistical representations. The houses attract attention, while their proportional bodies, numerical labels, and repeated colours keep that visual interest connected to the analytical purpose. The figure supports exploration and comparison without presenting regional associations as evidence of causation.

4 Design evolution and validation

The visualization was developed through an iterative process combining critical design assessment, exploration of alternative representations, and feedback from people outside the project. Throughout its development, we evaluated how effectively the graphical choices supported our research question,

progressively refining the design to balance creativity, analytical relevance, and accessibility to a broader audience.

4.1 Evolution and validation of the geographical visualization

Initial version: establishing the visual concept. The first prototype introduced the central concept of combining a geographical map with house-shaped symbols representing rental affordability. Each house was connected to its corresponding NUTS III region through a thin line, with its filled proportion indicating the share of mean gross monthly earnings required to rent an 80 m^2 dwelling. Exact percentage labels complemented this representation, enabling precise regional comparisons.

Initially, the map remained visually neutral, as our priority was to establish the layout and develop the house metaphor. Through internal evaluation, however, we recognized that the absence of geographical colour encoding limited the identification of spatial patterns. Although individual values were available, understanding the overall distribution required readers to examine the symbols and percentages separately. We therefore decided that the map should communicate the rent-to-earnings ratio directly, allowing regional patterns to emerge before readers explored individual values.

Intermediate version: enhancing geographical perception. To address this limitation, the second version introduced a continuous sequential colour scale, with progressively darker shades of red representing higher rent-to-earnings ratios. This additional visual encoding made geographical differences more immediately perceptible, while the house symbols continued to provide precise individual comparisons. The two representations therefore complemented each other, supporting both an overview of spatial patterns and a more detailed examination of regional values.

Nevertheless, further evaluation suggested opportunities to strengthen the communication of our main finding. Alongside our own assessment, we shared this version with people outside the project to gather feedback on its clarity and ease of interpretation. This provided an external perspective, helping us evaluate the design from the standpoint of viewers less familiar with the data. The process reinforced the importance of making the 50% threshold more visually prominent and improving the accessibility of the house-based representation. Although the continuous scale preserved gradual differences between regions, it did not sufficiently distinguish the threshold central to the main finding.

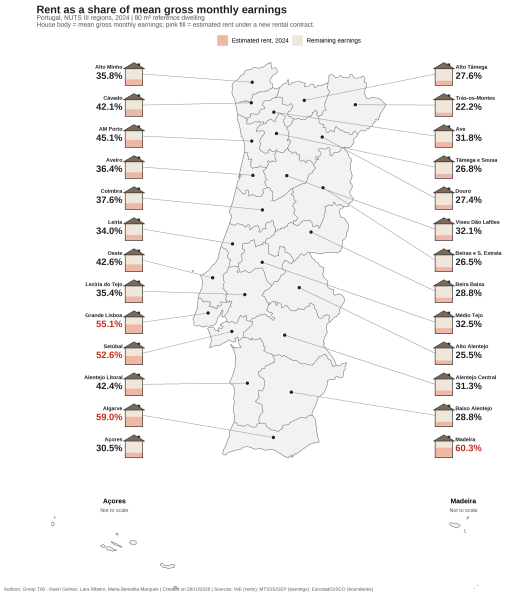


Figure 4. Initial prototype combining geographical boundaries with house-shaped symbols, without colour encoding of the regions.

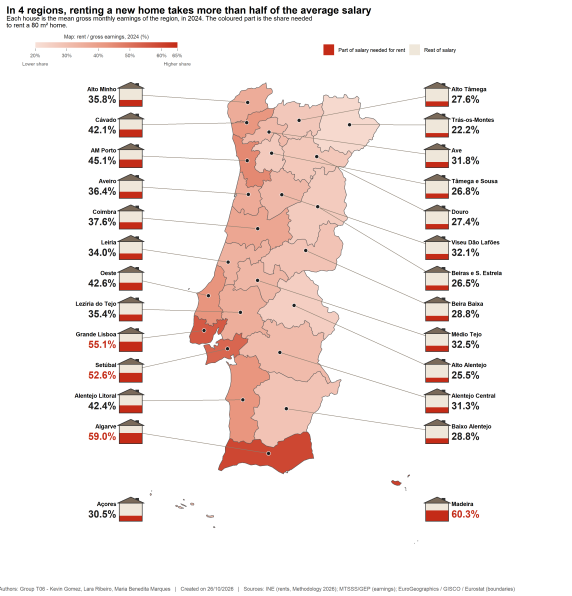


Figure 5. Intermediate prototype incorporating a continuous sequential colour scale to communicate regional differences in rent-to-earnings ratios.

Final version: refinement through evaluation and user feedback. Building on these observations, the final version introduced several coordinated refinements aimed at strengthening visual hierarchy and facilitating interpretation. These decisions combined our own critical assessment with external feedback, preserving the original concept while improving its effectiveness for a broader audience.

The most significant change was replacing the continuous colour scale with four discrete intervals: up to 30%, above 30% up to 40%, above 40% up to 50%, and over 50%. These categories provide easily interpretable ranges, with the darkest red reserved for regions where estimated rent exceeds half of mean earnings. This makes the four regions with the highest ratios immediately identifiable. Although discretization reduces the precision conveyed through colour, exact percentage labels preserve the quantitative detail. The house symbols were also assigned the same colour categories as the map, reinforcing consistency between the two representations.

We introduced the *How to read a house* guide to reinforce the interpretation of the house symbols, illustrating the relationship between the filled proportion, mean earnings, and the dashed 50% reference line. This concise visual explanation supports faster interpretation while strengthening the connection between the graphical encoding and the research question.

Finally, Açores and Madeira were reorganized into compact inset maps, identified as not being to scale. This improved the use of space and the overall visual balance while preserving the geographical inclusion of both archipelagos. Additional refinements to labels and legends contributed to a clearer reading sequence.

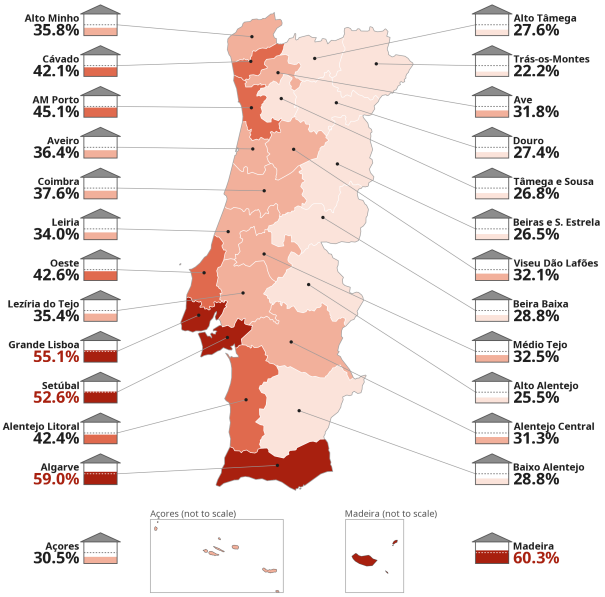


Figure 6. Final geographical visualization featuring discrete rent-to-earnings categories, consistent house-symbol encoding, explanatory guidance, and refined island insets.

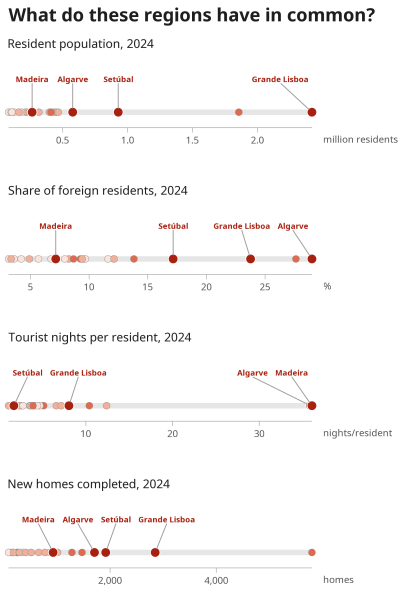


Figure 7. Final contextual dot plots; regions above the 50% threshold are highlighted.

Overall, the evolution of the map illustrates the importance of continuous design evaluation, complemented by external feedback. While our own assessment guided development throughout the project, feedback from other viewers provided an additional perspective on the effectiveness of our choices. This process strengthened the alignment between the intended message and its interpretation without compromising the original creative concept.

4.2 Development of the complementary visualizations

While the geographical visualization evolved through successive refinements of a consistent concept, the complementary charts followed a more exploratory design process. To address the second part of our research question, we considered different variables and graphical representations to determine which

regional characteristics could provide meaningful context for the areas with the highest rent-to-earnings ratios.

Throughout this exploration, we assessed each indicator according to its analytical relevance, contribution to understanding regional patterns, and suitability for comparison. This led us to prioritize four complementary dimensions: resident population, the share of foreign residents, tourist nights per resident, and newly completed dwellings. Together, these indicators provide distinct perspectives on demographic concentration, international population presence, tourism intensity, and housing supply.

We ultimately selected four aligned dot plots because their consistent structure facilitates comparisons through position along quantitative scales while remaining compact enough to integrate alongside the map. By highlighting the four regions where the rent-to-earnings ratio exceeds 50% and retaining the remaining regions as a reference, the charts allow readers to explore whether the highlighted regions share similar characteristics or exhibit contrasting profiles.

5 Final visualization and findings

Figure 1 combines the geographical distribution of rental affordability with four contextual indicators. The highlighted regions are those where the estimated monthly rent of an 80 m^2 dwelling exceeds 50% of regional mean gross monthly earnings.

5.1 Regional differences in rental affordability

Four regions exceed the 50% threshold: Madeira (60.3%), Algarve (59.0%), Grande Lisboa (55.1%), and Península de Setúbal (52.6%), labelled Setúbal in the figure. These are the least affordable regions under the project's common dwelling-size and earnings assumptions.

Terras de Trás-os-Montes records the lowest ratio, at 22.2%. The difference between the highest and lowest regional values is therefore 38.1 percentage points. The Porto Metropolitan Area reaches 45.1%, below the highlighted threshold but still representing a substantial share of mean gross monthly earnings.

The map reveals a concentration of high ratios around the Lisbon metropolitan area, in Algarve, and in Madeira. However, the pattern is not a simple division between coastal and inland regions: the exact labels show variation within these broader geographical groups.

5.2 Shared characteristics and contrasting profiles

The contextual panels show that the four highlighted regions do not share a uniform profile across all indicators.

Grande Lisboa has the largest resident population among the 26 regions, whereas Setúbal, Algarve, and Madeira have substantially smaller populations. Population size alone therefore does not distinguish all four high-ratio regions from the remaining territories.

The foreign-resident panel shows a clearer similarity between Algarve, Grande Lisboa, and Setúbal, which appear towards the upper end of the regional distribution. Madeira has a lower foreign-resident share, despite recording the highest rent-to-earnings ratio. A comparatively high foreign-resident share is therefore shared by three highlighted regions, but not by the entire group.

Tourism reveals another contrast. Algarve and Madeira lie at the upper extreme of tourist nights per resident, while Grande Lisboa and, particularly, Setúbal have lower values. High rent-to-earnings ratios thus coexist with different levels of tourism intensity.

In the construction panel, Grande Lisboa has the highest number of newly completed dwellings among the four highlighted regions, followed by Setúbal, Algarve, and Madeira. None records the highest total across all 26 regions. Because this indicator is an absolute count, these differences cannot establish whether construction was sufficient relative to regional population, housing stock, or demand.

Together, the panels identify overlapping characteristics rather than a single profile shared by all four regions. Algarve combines a high foreign-resident share with high tourism intensity; Grande Lisboa

combines a high foreign-resident share with a large resident population; Setúbal has a comparatively high foreign-resident share but low tourism intensity; and Madeira combines the highest affordability ratio with high tourism intensity and a lower foreign-resident share. These are descriptive regional comparisons, not estimates of the effects of these characteristics on rents.

6 Limitations and conclusion

6.1 Limitations

The rent-to-earnings ratio provides a standardized comparison across regions, based on an 80 m² reference dwelling, rents from new contracts, and mean gross monthly earnings. It should therefore be interpreted as an indicator of regional rental affordability rather than the actual share of income spent on housing by individual households. The 50% threshold serves as a descriptive reference within the project.

Regional values summarize conditions within each NUTS III territory and may conceal differences between municipalities and population groups. The contextual indicators also require distinct interpretations: for example, newly completed dwellings measure construction volume rather than its adequacy relative to housing demand.

Finally, the contextual panels describe regional patterns in 2024. They help identify similarities and differences between the highlighted regions, but do not establish causal relationships between these characteristics and rental affordability.

6.2 Conclusion

Under the project's reference-dwelling and earnings assumptions, Madeira, Algarve, Grande Lisboa, and Península de Setúbal are the least affordable regions to rent in, with estimated rent exceeding half of regional mean gross monthly earnings.

These regions share some characteristics but do not form a uniform group. Algarve, Grande Lisboa, and Setúbal have comparatively high foreign-resident shares, while Algarve and Madeira stand out for tourism intensity. Population size and construction volumes vary substantially within the group.

By combining a map, proportional house indicators, and contextual distributions, the visualization answers where the highest ratios occur and shows how the highlighted regions resemble or differ from one another. It provides an accessible regional comparison while leaving causal explanations and household-level affordability assessment to further analysis.

7 Team contributions and use of generative AI

7.1 Team contributions

The project was developed collaboratively, with all three team members actively participating in discussions, design decisions, and the refinement of the final visualization, rather than following a strict division of tasks.

Kevin Gomez was primarily responsible for collecting, cleaning, and preprocessing the data, integrating the different sources into the final dataset used for the analysis. He also contributed to the development and refinement of the visualizations, particularly through the joint evaluation of the map and the exploration of complementary charts.

Lara Ribeiro proposed the initial visualization concept and developed the first prototype of the geographical map, establishing the foundation for its subsequent iterations. She also contributed to refining the map based on internal assessment and external feedback, as well as exploring and developing the complementary visualizations.

Maria Benedita Marques contributed to the iterative improvement of the geographical visualization and participated in the exploration, development, and evaluation of the complementary charts. She was also responsible for organizing and structuring the final report and converting it from Word to LaTeX, ensuring consistency in its structure and presentation.

Additionally, all three members contributed to writing the report, developing alternative versions of the complementary charts, and discussing the final design choices. The geographical visualization was progressively refined through joint contributions, incorporating improvements proposed by each member throughout the process.

7.2 Use of generative AI

Generative AI was used as a supplementary tool to assist with specific technical and writing-related tasks throughout the project. In particular, it provided occasional guidance on functionalities beyond those directly covered in class, such as handling geographical coordinates and boundaries to establish the base geographical representation of Portugal, as well as implementing specific graphical details of the house symbols, including their geometric shapes and proportional filling. It also supported debugging selected sections of the R code, translating and refining the English wording of parts of the report, and adjusting formatting during the transition from Word to LaTeX.

8 References

[1] Agência para a Integração, Migrações e Asilo. Relatórios de migrações e asilo. AIMA. Official publication portal. URL https://aima.gov.pt/pt/a-aima/relatorios-de-migracoes-e-asilo.

[2] European Commission, Eurostat/GISCO. NUTS 2024: Geographical datasets. GISCO data distribution service. URL https://gisco-services.ec.europa.eu/distribution/v2/nuts/nuts-2024-files.html.

[3] Tiffany France. Choosing fonts for your data visualization. Nightingale, Data Visualization Society, June 2020. URL https://nightingaledvs.com/choosing-fonts-for-your-data-visualization/. Published 8 June 2020.

[4] HCL Wizard. Colour-vision deficiency emulator. Online colour-vision simulation tool. URL http://hclwizard.org:3000/cvdemulator/.

[5] Instituto Nacional de Estatística. Statistical database. Statistics Portugal. Online statistical database. URL https://www.ine.pt/xportal/xmain?xpid=INE&xpgid=ine_base_dados.